

JPEG AI as a Threat to Deepfake Detection

Computer Vision | Sapienza University of Rome

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Outline

01

Problem Statement

The emergence of learned neural image codecs (JPEG AI)

02

Forensic Analysis

How JPEG AI compression modifies the low-level statistical fingerprints that deepfake detectors exploit.

03

Proposed Methodology

Mitigation strategies that make detector more robust to deepfake.

04

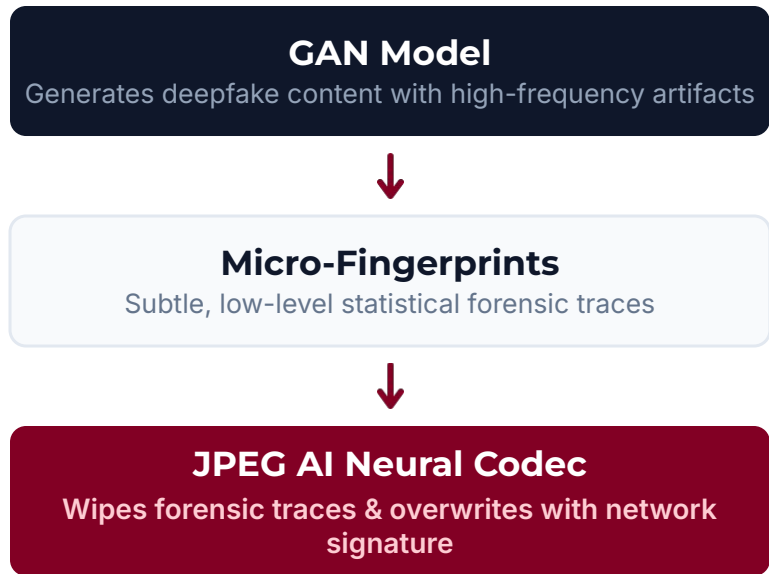
Experimental Evaluation

Degradation curves and mitigation strategy benchmarks.

Problem Statement

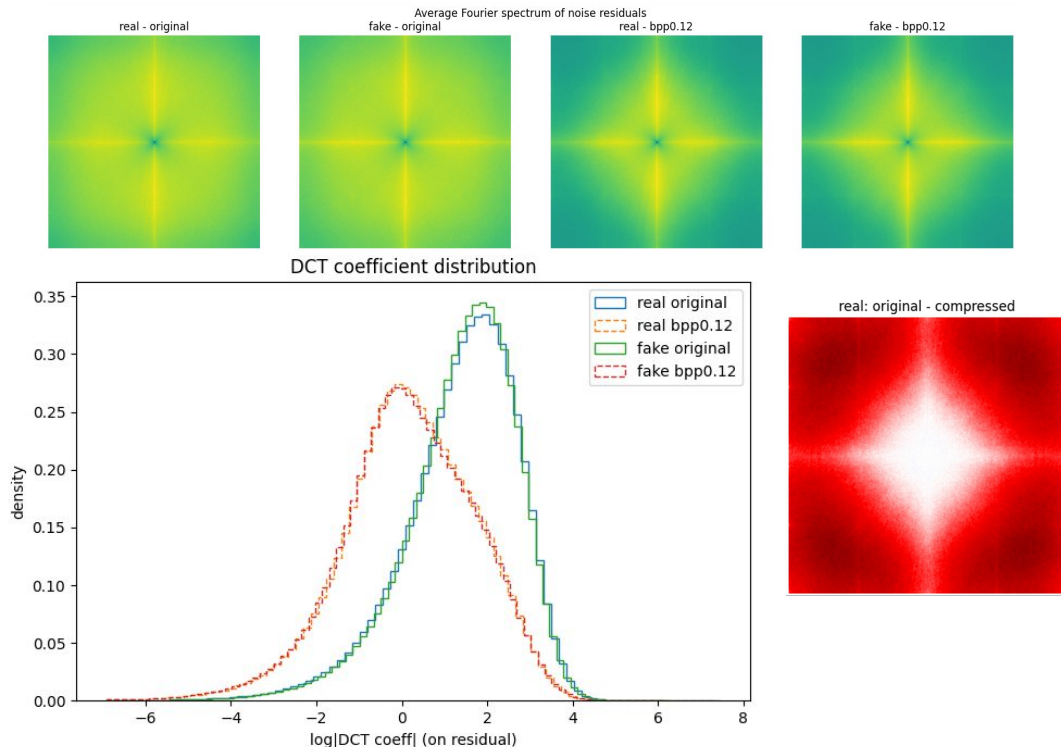
Key Context

- Deepfake detectors rely on **subtle, low-level statistical fingerprints** left in high-frequency content during generation.
- Any operations that smooth or overwrite these pixel-level traces can completely erase the evidence detectors depend on.
- **JPEG AI Standard:** The first international standard for end-to-end learned compression driven by deep neural networks.
- **Core Hypothesis:** Unlike traditional JPEG, JPEG AI neural encoding completely overwrites native artifacts with its own network signature, neutralizing forensic models.



Problem Statement: Forensic Analysis

- **High-Frequency Tracking:** Analysis is performed on high-pass noise residuals to expose hidden, low-level generative fingerprints.
- **2D Fourier Star Pattern:** Neural compression forces a distinct, highly directional "**bright cross**" onto the pixel spectrum, signaling a codec stamp.
- **DCT Magnitude Suppression:** Codecs aggressively eliminate fine details, crushing the statistical distribution of DCT coefficients toward zero.
- **Fingerprint Erasure:** The difference map (*original – compressed*) isolates the exact high-frequency energy that JPEG AI strips away — the bright central core marks the fingerprint the codec removes.



State of the Art

CNNDetection Detector (Wang et al., CVPR 2020)

A ResNet50-based model built on the premise that CNN-generated images share common architectural artifacts.

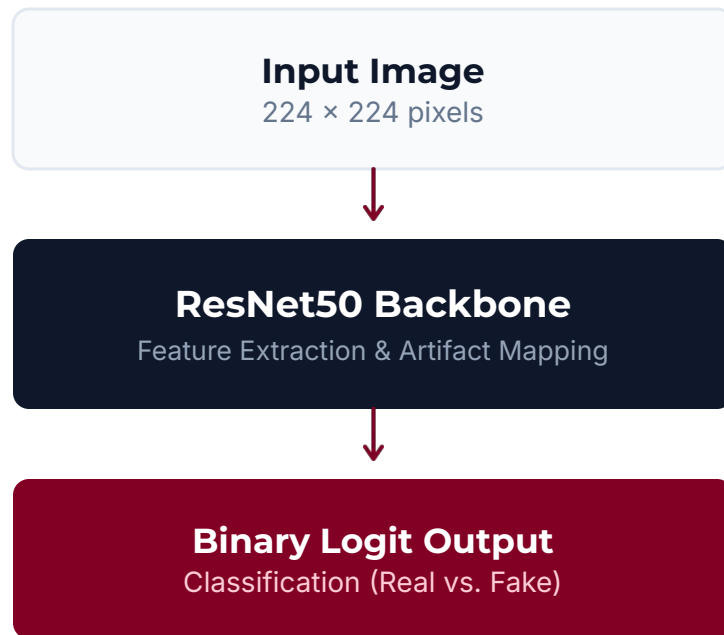
Evaluated via two training configurations with an augmentation of Gaussian blurring and standard JPEG compression:

- **prob0.1**: highly vulnerable baseline with light augmentation.
- **prob0.5**: robust baseline featuring heavy augmentation.

The Core Research Gap

The forensic vulnerabilities induced by the newly established neural standard JPEG AI remain almost entirely unmapped in active literature.

Our project reproduces the severe degradation challenges introduced by *Cannas et al. (ICCVW 2025)* while constructing ad-hoc mitigation strategies.



[1] Wang, S.-Y., Wang, O., Zhang, R., Owens, A., & Efros, A. A. (2020). CNN-generated images are surprisingly easy to spot... In *Proceedings of the IEEE/CVF CVPR*.

[2] E. D. Cannas, P. Bestagini, S. Tubaro, et al., "Is JPEG AI Going to Change Image Forensics?", IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), 2025.

Proposed Method

01

Fine-Tuning: Augmentation

Retrains the detector on a 50/50 mix of original and JPEG-AI-compressed images.

Creates two bitrate-specialized experts: low (**0.12–0.25 bpp**) and high (**0.50–0.75 bpp**) to combat codec loss.

JPEG AI Compression



50/50 Dataset (Orig + Comp)



Fine-Tune & Experts

02

Fine-Tuning: Consistency

Adds an invariance loss on original-compressed pairs, forcing matching scores:

$$\text{BCE}(\text{orig}) + \text{BCE}(\text{comp}) + \lambda \cdot \text{MSE}(\sigma(\text{orig}), \sigma(\text{comp}))$$

JPEG AI Compression



Paired Dataset (Orig ↔ Comp)



Consistency Loss Fine-Tune

03

Test-Time Sharpening

Test-time preprocessing only, requiring no neural network retraining.

Applies an unsharp mask before detection to restore high-frequency traces smoothed away by the JPEG AI codec.

JPEG AI Compression



At Inference: Sharpen Image



Standard Detector

Trade-off: Strategies **01** and **02** shift computational cost to a one-time offline training (zero run-time overhead). Strategy **03** avoids training entirely but increases inference latency and degrades detection accuracy.

Dataset

Sources & Format

Core Benchmark: ProGAN dataset (extracted from the official CNNDetection testing splits).

Image Sources: Real photographs from LSUN alongside synthetic images generated via ProGAN.

Resolution: Uniformized at a native resolution of 256×256 pixels.

Data Isolation

Strict isolation designed to prevent leakage across experimental subsets.

- **Evaluation Split:** 150 images/class (300 total) per target bitrate.
- **Fine-Tuning Split:** 100 images/class, carved from a disjoint index range separate from test data.

Infrastructure

Storage & Sync: Handled dynamically via cloud integration.

Google Drive Mounts: Enable seamless execution and consistent environment synchronization across processing nodes.

Experimental Setup

01

Neural Codec Configuration

Executed using the official JPEG AI reference software under standard profiles (base.json) with secondary tools deactivated (tools_off.json). Used 4 bpp: (0.12, 0.25, 0.50, 0.75)

02

Crucial Preprocessing

Enforced strict deterministic **CenterCrop(224)** with **no resizing**. This preserves the raw spatial layout and micro-pixel structures required for artifact verification.

03

Optimization Settings

Retrained across 8 full epochs, utilizing a conservative Learning Rate of $5e-5$ paired with the Adam optimizer.

04

Multi-Expert Architecture

Formulated two specialized models: a **Low-Bitrate Expert** (optimized for bpp 0.12–0.25) and a **High-Bitrate Expert** (optimized for bpp 0.50–0.75).

Baseline Evaluation

Key Observations

Perfect Baseline: On uncompressed original images, the detector achieves perfect classification with **AUC = 1.000** and **Accuracy = 1.000**.

Severe Degradation: Increasing neural compression strength causes a **monotonic, severe performance drop** across all metrics.

Critical Failure Point: At maximum compression (b_{pp} 0.12), performance drops sharply: AUC falls to **0.781** and Accuracy plummets to **68.3%**.

Forensic Failure: The False Positive Rate leaps to **0.480**, misclassifying nearly half of the genuine images as fakes due to neural artifacts.

prob0.1

CONDITION	AUC	AP	Acc	FPR	FNR	F1	n
original	1.000	1.000	1.000	0.000	0.000	1.000	300
bpp 0.12	0.707	0.667	0.577	0.787	0.060	0.689	300
bpp 0.25	0.873	0.851	0.683	0.607	0.027	0.755	300
bpp 0.50	0.976	0.980	0.853	0.267	0.027	0.869	300
bpp 0.75	0.991	0.992	0.940	0.093	0.027	0.942	300

prob0.5

CONDITION	AUC	AP	Acc	FPR	FNR	F1	n
original	1.000	1.000	1.000	0.000	0.000	1.000	300
bpp 0.12	0.781	0.753	0.683	0.480	0.153	0.728	300
bpp 0.25	0.944	0.944	0.860	0.233	0.047	0.872	300
bpp 0.50	0.996	0.997	0.960	0.060	0.020	0.961	300
bpp 0.75	0.999	0.999	0.977	0.020	0.027	0.977	300

Evaluation: Mitigation Benchmarks

Fine-Tuning: Augmentation

Effectively reclaims robustness at the critical bpp 0.12 boundary, boosting **AUC back to 0.875** and **Accuracy to 76.3%** while forcing FPR down to 0.340.

Fine-Tuning: Consistency

Delivers solid, balanced recovery (**AUC = 0.864**, **Acc = 76.0%**) while achieving the lowest overall false alarm rate with an **FPR of 0.320**.

Test-Time Sharpening

Preprocessing images via Unsharp Masking actively worsens detection, reducing AUC further down to **0.742**. Naive high-frequency contrast enhancement cannot restore structural features deleted by neural models.

High-Bitrate Convergence: At high bitrates (bpp ≥ 0.50), where the original baseline is already strong, both training frameworks smoothly converge to near-perfection (**AUC ≥ 0.999**).

Evaluation: Mitigation Benchmarks B

CNNDetection — prob0.1

bitrate	strategy	AUC	AP	Acc	BalAcc	FPR	FNR	F1	EER
original	Base	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000
	Fine-tuning	1.000	1.000	0.997	0.997	0.000	0.007	0.997	0.007
	Consistency	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000
bpp 0.12	Base	0.707	0.667	0.577	0.577	0.787	0.060	0.689	0.327
	Fine-tuning	0.811	0.823	0.737	0.737	0.300	0.227	0.746	0.267
	Consistency	0.803	0.808	0.660	0.660	0.547	0.133	0.718	0.260
bpp 0.25	Base	0.873	0.851	0.683	0.683	0.607	0.027	0.755	0.193
	Fine-tuning	0.937	0.939	0.853	0.853	0.180	0.113	0.858	0.140
	Consistency	0.947	0.951	0.813	0.813	0.327	0.047	0.836	0.127
bpp 0.50	Base	0.976	0.980	0.853	0.853	0.267	0.027	0.869	0.073
	Fine-tuning	0.973	0.963	0.927	0.927	0.120	0.113	0.930	0.067
	Consistency	0.984	0.980	0.903	0.903	0.180	0.013	0.911	0.053
bpp 0.75	Base	0.991	0.992	0.940	0.940	0.093	0.027	0.942	0.047
	Fine-tuning	0.986	0.977	0.953	0.953	0.073	0.020	0.955	0.040
	Consistency	0.994	0.990	0.957	0.957	0.087	0.000	0.958	0.033

CNNDetection — prob0.5

bitrate	strategy	AUC	AP	Acc	BalAcc	FPR	FNR	F1	EER
original	Base	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000
	Fine-tuning	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000
	Consistency	1.000	1.000	1.000	1.000	0.000	0.000	1.000	0.000
bpp 0.12	Base	0.781	0.753	0.683	0.683	0.480	0.153	0.728	0.260
	Fine-tuning	0.875	0.884	0.763	0.763	0.340	0.133	0.785	0.193
	Consistency	0.864	0.873	0.760	0.760	0.320	0.160	0.778	0.207
bpp 0.25	Base	0.944	0.944	0.860	0.860	0.233	0.047	0.872	0.133
	Fine-tuning	0.974	0.968	0.897	0.897	0.173	0.033	0.903	0.060
	Consistency	0.971	0.972	0.910	0.910	0.133	0.047	0.914	0.053
bpp 0.50	Base	0.996	0.997	0.960	0.960	0.060	0.020	0.961	0.027
	Fine-tuning	0.999	0.999	0.977	0.977	0.040	0.007	0.977	0.013
	Consistency	0.999	0.999	0.987	0.987	0.020	0.007	0.987	0.013
bpp 0.75	Base	0.999	0.999	0.977	0.977	0.020	0.027	0.977	0.020
	Fine-tuning	1.000	1.000	0.987	0.987	0.027	0.000	0.987	0.013
	Consistency	1.000	1.000	0.983	0.983	0.027	0.007	0.983	0.007

Notice that for fine-tuning detectors we reported only results of “low” type (Augmentation: [0.12, 0.25]), that are more effective than “high” (Augmentation: [0.50, 0.75]).

Conclusions & Future Work

Conclusions

JPEG AI Threat: Compression poses a severe, quantifiable threat to modern multimedia forensics by wiping low-level GAN high-frequency fingerprints.

Naive Failures: Post-processing solutions at test-time (e.g. Sharpening) fail entirely since they cannot hallucinate destroyed statistics.

Robust Recovery: Retraining models via balanced augmentation (Fine-Tuning) and output-level probability mapping recovers a major portion of the forensic gap.

Future Work

Limitations & Next Steps: Current evaluations are restricted to In-Distribution scenarios where ProGAN is used for both training and testing.

Cross-Generator Datasets: Future iterations require validation against completely disjoint cross-generator datasets to confirm generalization.

DANN Deployment: Implement Domain-Adversarial Neural Networks (DANN) to achieve complete, compression-invariant detection.

References

[1] E. D. Cannas, P. Bestagini, S. Tubaro, et al., "Is JPEG AI Going to Change Image Forensics?", IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), 2025.

[2] P. Wang, O. Wang, R. Zhang, A. Owens, A. A. Efros, "CNN-generated images are surprisingly easy to spot... for now", IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[3] JPEG AI Reference Software Official Codebase, ISO/IEC JTC 1/SC 29/WG 1.
